

Looking Ahead in Goal-Oriented Dialogue

Preference-Tree (PTO) vs Group-Relative (GRPO) optimization of a small Motivational-Interviewing therapist

Project status — all four arms, both graders, and a new look-ahead result

Both comparisons are finished. The look-ahead arm reached iteration 10 this week and resolved — not the way we last read it.

Lior Baruch · Reichman University · 16 August 2026

Llama-3.2-1B therapist (bf16) · gpt-4o-mini simulated patient + oracle · Claude Haiku 4.5 held-out judge · 96 personas, persona-paired

Where the project stands, and what I'd like to decide

Settled

- **The main comparison is finished.** PTO and GRPO, 10 iterations each at matched settings, every conversation scored on eight instruments by two independent graders.
- **The instrument is measured, not assumed.** Oracle repeatability, a full re-scoring by a judge from another model family that never played the patient, and an explicit list of where the two graders disagree.
- **The look-ahead arm is finished too.** PTO $K=5$ now trains to iteration 10 and is scored 0–10 on both graders. It closes the reward-hack channel it targets for the whole run, and the therapist substitutes another violation for the one it lost. That is new this week.
- **Two paper drafts exist**, on deliberately disjoint claims, both compiling.

Open — the agenda

- **Scope.** Two drafts, one thesis. Which is the chapter, which is the paper, and in what order do they get finished?
- **Budget.** Two things are left to buy: GRPO at $K=5$, or nothing. Only the first changes a claim.
- **Human MI coders.** The one validity gap that money alone does not close, and the one a reviewer will ask about first.
- **Authorship and venue** for both drafts.

In one line: the therapist learns to flatter; a principled fix stops the flattery and the therapist starts lecturing instead, with no fall in total MI violations that a grader outside the loop can see. The reward function cannot tell any of these apart.

The starting point — the ICLR 2025 paper

Preference Tree Optimization: Enhancing Goal-Oriented Dialogue with Look-Ahead Simulations

Baruch, Butman, Bar, Friedman · ICLR 2025 (SSI-FM workshop poster)

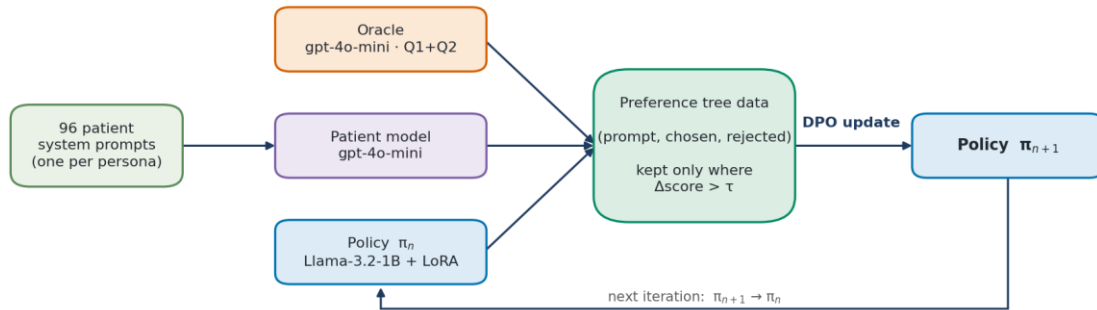
- **The method.** At each therapist turn, branch several candidate replies; for each, simulate **K further turns**; let the oracle score the resulting trajectory; keep best and worst as a preference pair; update with DPO. Repeat, regenerating from the improved model.
- **Why look ahead.** Scoring a reply on its own rewards openings that look good in isolation. Scoring the reply plus K simulated turns rewards openings that lead somewhere.
- **Headline finding.** Every PTO model beat the untrained baseline, and **K = 5 scored higher and more stably than K = 0** — on a 7B therapist over 7 iterations.
- **Why that matters today.** This experiment does not reproduce that on a 1B model over 10 iterations, and the reason turns out to be interesting rather than a contradiction. Slides 11–14.

What was run, and what is scored

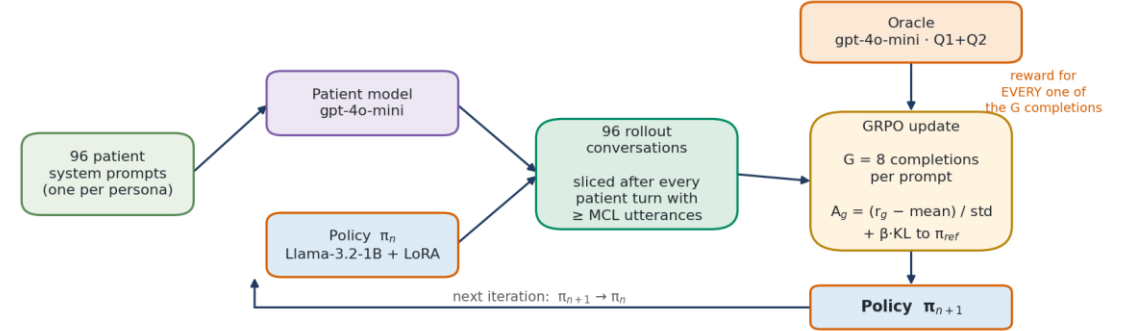
Arm	Trained	Scored on BOTH graders	Status
PTO K=0	iterations 1–10	base + 1–10	complete
GRPO K=0	iterations 1–10	base + 1–10	complete
PTO K=5	iterations 1–10	base + 1–10	complete
GRPO K=5	iteration 1	base + 1	thin — stopped on budget

- **35 scored model states**, each 96 conversations, each conversation scored on eight instruments by two graders.
- **Matched by construction** across every arm: sampling temperature 1.2, 8 candidates per branch point, minimum context length 12, the same τ filter, the same DPO β , the same oracle. Verified against each run's stored metadata, not assumed.
- **The same 96 patient personas recur in every iteration** — reshuffled, so identity has to be recovered rather than read off file order. That is what makes every comparison in this deck **persona-paired**, which is far stronger than comparing group means.
- **Training reward = two rubrics** (session satisfaction + working alliance). The other six instruments were never optimised against, which is what makes them evidence.

The two methods — same loop, one step different



Green = data this iteration produces · purple = an API model we call · blue = the policy being trained · orange = the oracle
 The 96 conversations generated at the start of the iteration are also the evaluation set for π_n — there is no separate evaluation pass.



Same colour code as the PTO framework figure. The one structural difference is where the oracle sits: inside the update, not before it.
 As in PTO the rollout conversations double as the evaluation set for π_n — but the prompt list is fixed for the iteration, every prompt trains, and nothing is discarded.

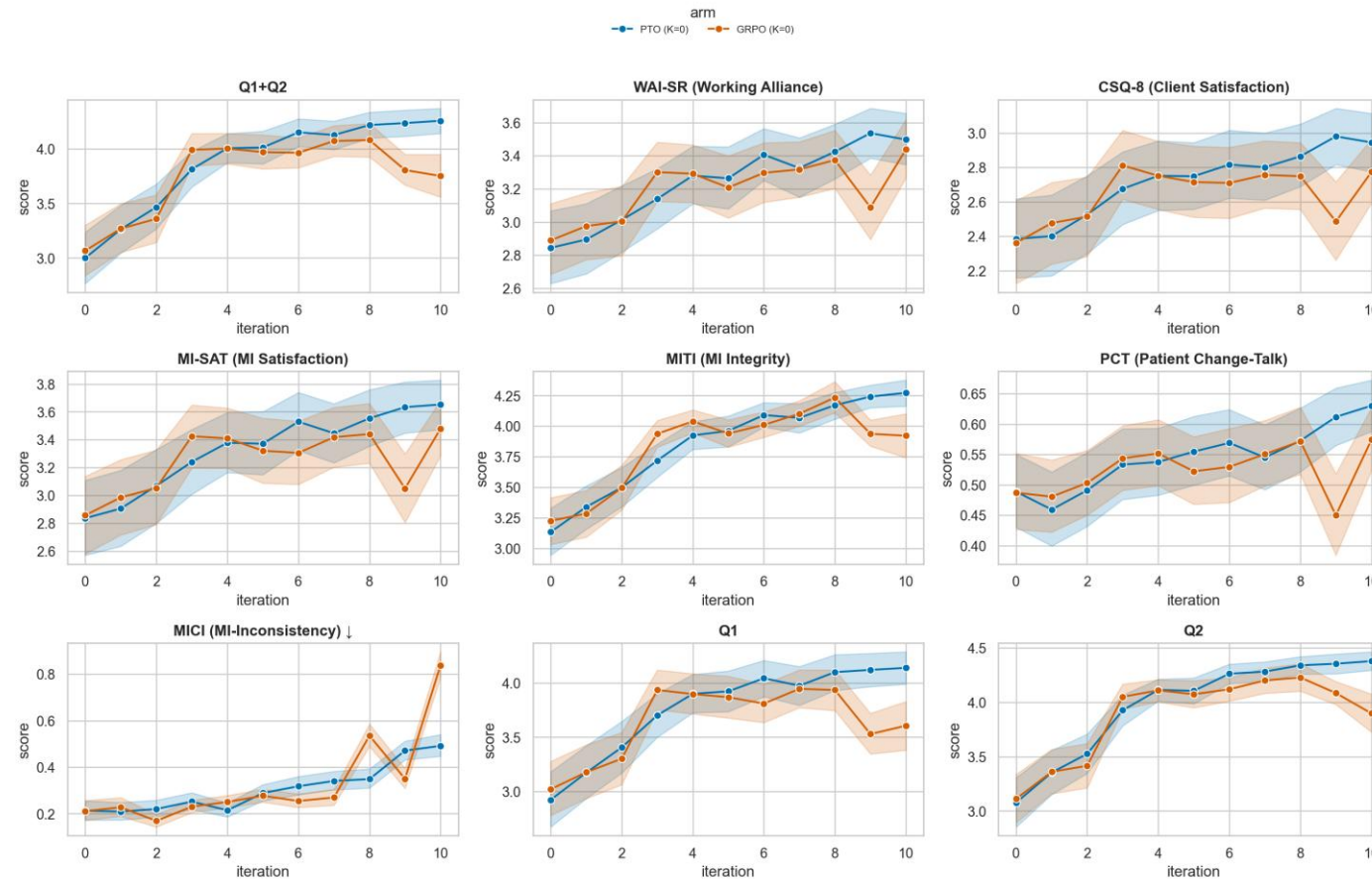
PTO — branch, score, keep best vs worst, DPO

GRPO — sample a group, standardize rewards, policy gradient

Both regenerate their own training data from the current policy each iteration. Look-ahead (K) changes only what CONTEXT the oracle grades — never the loss, the hyperparameters, or the sampling.

All eight instruments across 10 iterations (K = 0)

Outcome trajectories across iterations — full-conversation eval



The 9 metrics share one dominant global-evaluation (halo) factor (PC1=55%; technique + MI-inconsistency load a second factor, PC2=16%), so a uniform rise partly reflects one axis, not independent multi-skill gains.

Mean $\pm$ 95% CI over the 96 personas, primary oracle. MICI is lower-is-better. PTO climbs steadily to iteration 10; GRPO peaks at iteration 8 and regresses.

Endpoint numbers (K = 0)

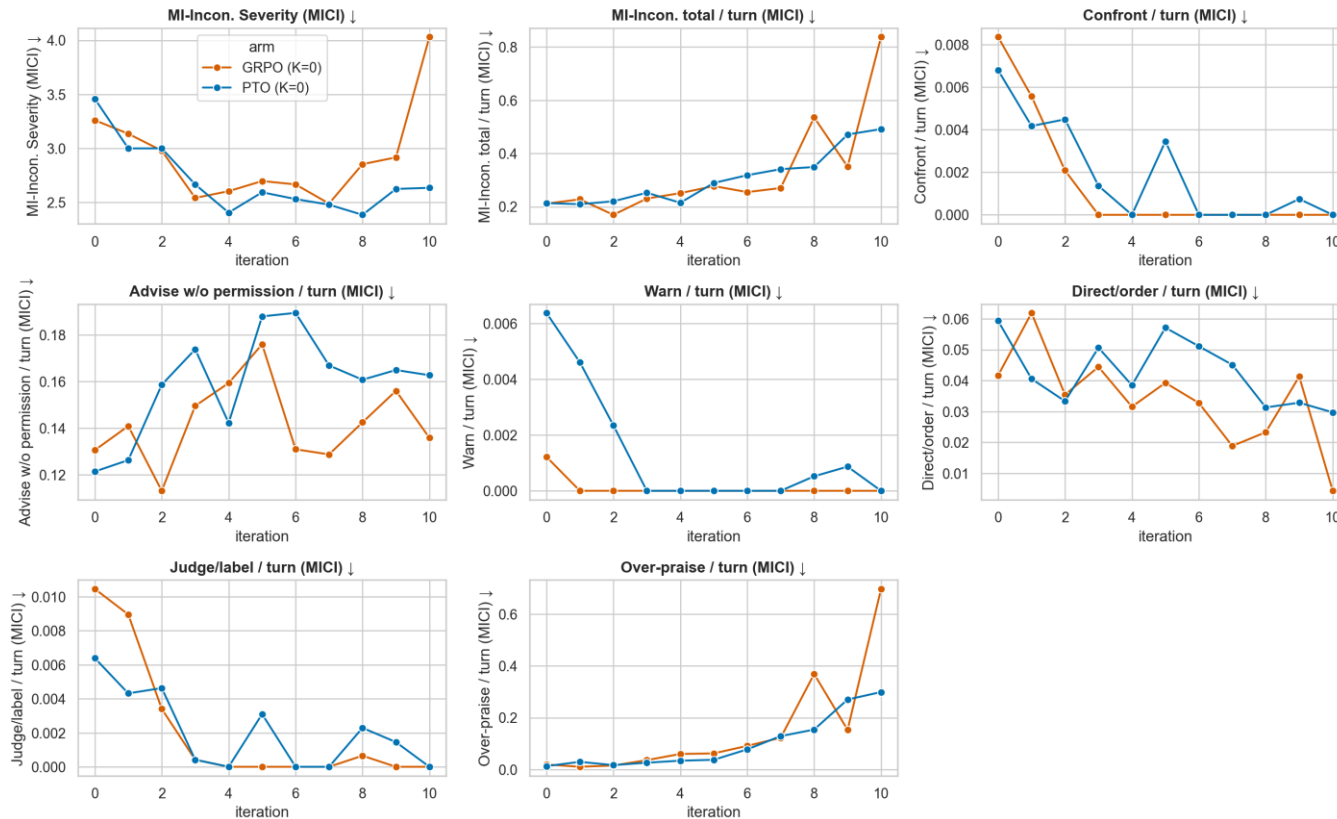
Primary oracle, final and best endpoint. Read off the generated scorecard (results/L0/tables/0_headline/gpt-4o-mini/) — every cell traces to the score lake.

	Q1+Q2	MITI	MICI ↓	R:Q	% MI-consistent
PTO K=0 @ iter 10	4.26	4.27	0.49	0.75	0.70
GRPO K=0 @ iter 10	3.75	3.92	0.84	1.44	0.83
GRPO K=0 @ iter 8 (its peak)	4.08	4.23	0.54	1.04	0.71

- **PTO beats GRPO at the matched endpoint** — Q1+Q2 4.26 vs 3.75, persona-paired +0.51, dz 0.73.
- **Credit GRPO at its peak instead** and PTO still leads (+0.18, dz 0.30, p = .010) — but the MITI and MICI gaps stop being significant. **We claim the weaker version.**
- **GRPO's higher R:Q is not a win.** Its reflection-to-question ratio crosses the MITI competency bar because it stopped asking questions — the regex question rate falls 0.83 → 0.15 per turn. A ratio is gameable through its denominator.
- **The held-out judge orders them the same way** on Q1+Q2 at iteration 10 — 2.87 (PTO) vs 2.26 (GRPO) — on a different absolute scale. **Never average the two graders' raw scores:** the primary one WAS the training reward, so this is train-vs-test, and the level offset is model-dependent. Only contrasts combine.

What the model actually learned (K = 0)

MICI detail — severity + per-therapist-turn rates (higher = worse)



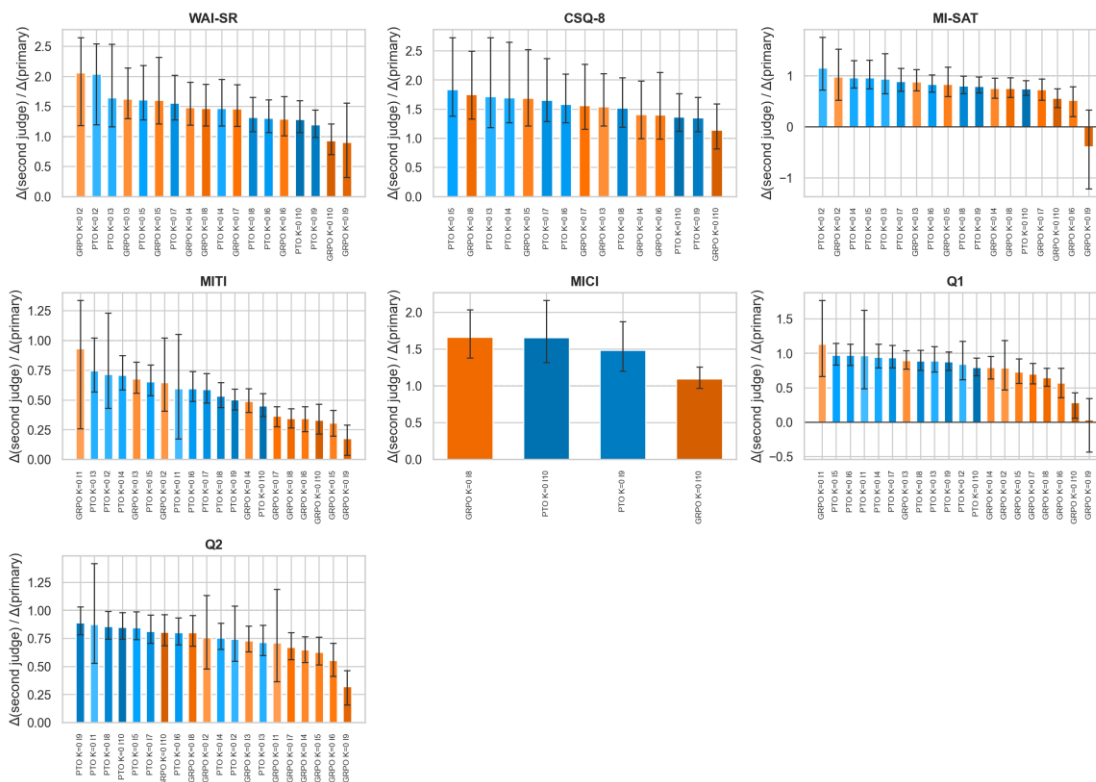
MI-inconsistent behaviour, decomposed, per therapist turn. Higher is worse.

One channel carries the whole thing

- Every **coercive** MI violation FALLS with training — confronting, warning, directing and judging all go to ~0.
- Over-praise rises from **0.02 to 0.70 per turn** in GRPO and 0.01 to 0.30 in PTO, and accounts for more than 100% of the increase in total MI-inconsistency.
- The rubrics measure how the session **felt to the client**. Flattery is the cheapest way to move that.
- So the model did not get broadly worse at MI. It swapped coercion for flattery.

How much of this survives a grader that took no part in training

[EVAL] Does the improvement transfer? Gain over PTO K=0 Base as seen by Claude Haiku 4.5, relative to the primary oracle it was trained against (dashed = full transfer)



Fraction of each arm's gain over base that survives the judge swap. ~ 1.0 = a real behaviour change; ~ 0 = a gain that existed only in the optimised grader.

Gain retention = a train/test split on the reward

- The primary oracle **was the training signal**; Claude Haiku 4.5 is held out, from another family, and never played the patient. Agreement between them is a generalisation check, not inter-rater reliability.
- Oracle repeatability ICC(2,1) **0.86–0.99**; 1,632 of 1,848 arm $\times$ metric contrasts keep their sign (98.9% at $|\Delta| \geq 0.50$).
- Q1 retention: PTO@10 **0.80** vs GRPO@10 **0.28**, non-overlapping — GRPO's late gains are increasingly grader-specific.
- ⚠ Two standing caveats: MITI dependability is **0.65** off one judge, and MICI agreement is weak at the conversation level. Both are limitations in the write-up, not footnotes.

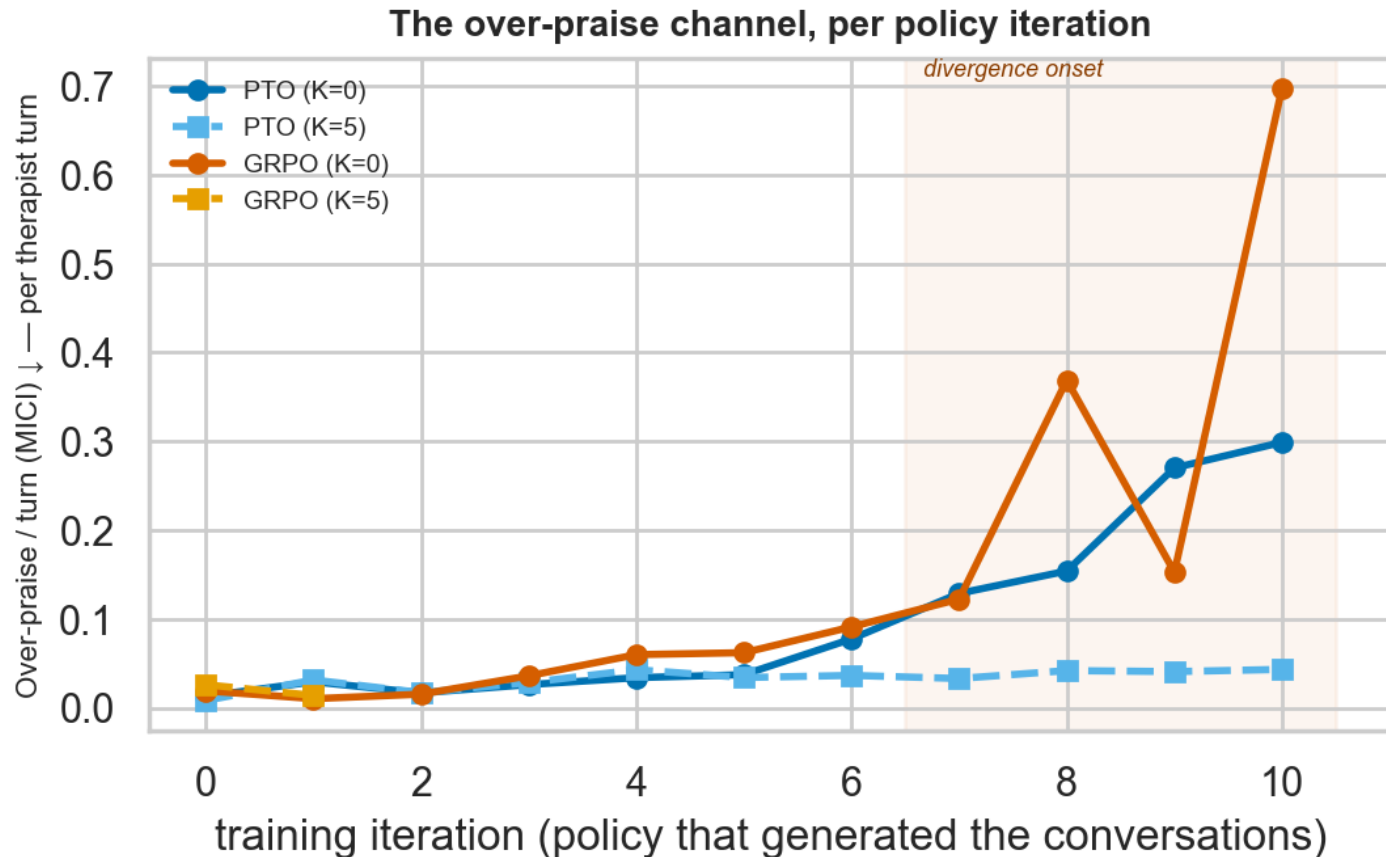
New this week

Does look-ahead fix the reward hack?

It closes the channel it targets. The therapist substitutes another violation for the one it lost.

PTO only, $K \in \{0,5\}$, eleven matched iterations (0–10), both graders, persona-paired.

Look-ahead closes the channel it targets, and keeps it closed



Over-praise per therapist turn, all four arms. K=0 solid, K=5 dashed. Shading marks the measured divergence onset. Higher is worse.

Exactly what it was supposed to do

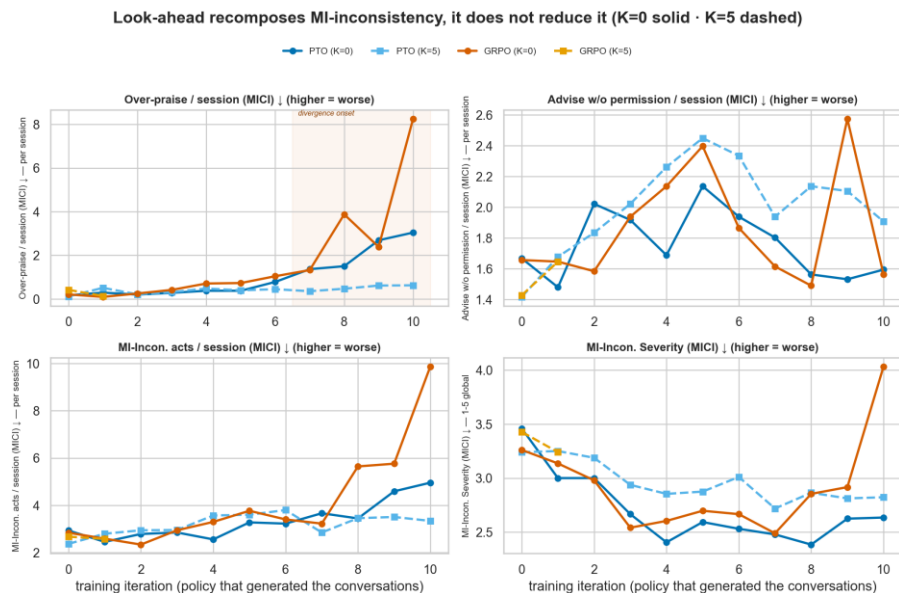
- The two PTO arms are **indistinguishable for six iterations** and then separate for good. Over-praise per session at iteration 10: **3.042 (K=0) vs 0.625 (K=5)**.
- Persona-paired, $n = 96$: **dz 0.887, $p < 10^{-4}$** . The held-out judge reports a larger gap in the same direction — 4.750 vs 1.177, dz 0.999.
- **Prevention, not delay.** K=5's over-praise stays LOW to the end of the run — it rises 0.115 $\rightarrow$ 0.625 across the eleven points, while K=0 goes 0.167 $\rightarrow$ 3.042. Low and still creeping, not static; the draft's open question is closed in the direction of prevention.
- Not a denominator artifact — these are **raw per-session counts**. K=5 actually takes MORE therapist turns (14.4 vs 10.2), which would push a rate the other way.
- Not a data-starvation artifact — over the full ten iterations K=5 trained on **more** preference groups (6,416 vs 4,935).
- And it is visible **upstream**, in what the reward itself selects for — not only in the output. Slide 13.

...and the substitution replicates — the reduction does not

per session, iteration 10	Δ (K=0 – K=5)	dz	p (Holm)
Over-praise	+2.417	+0.887	$< 10^{-4}$
Advice without permission	−0.313	−0.239	.057 n.s.
Directing the client	−0.375	−0.389	.0016
ALL MI-inconsistent acts	+1.615	+0.446	.0003

The therapist changed how it fails; whether it fails less depends on who grades

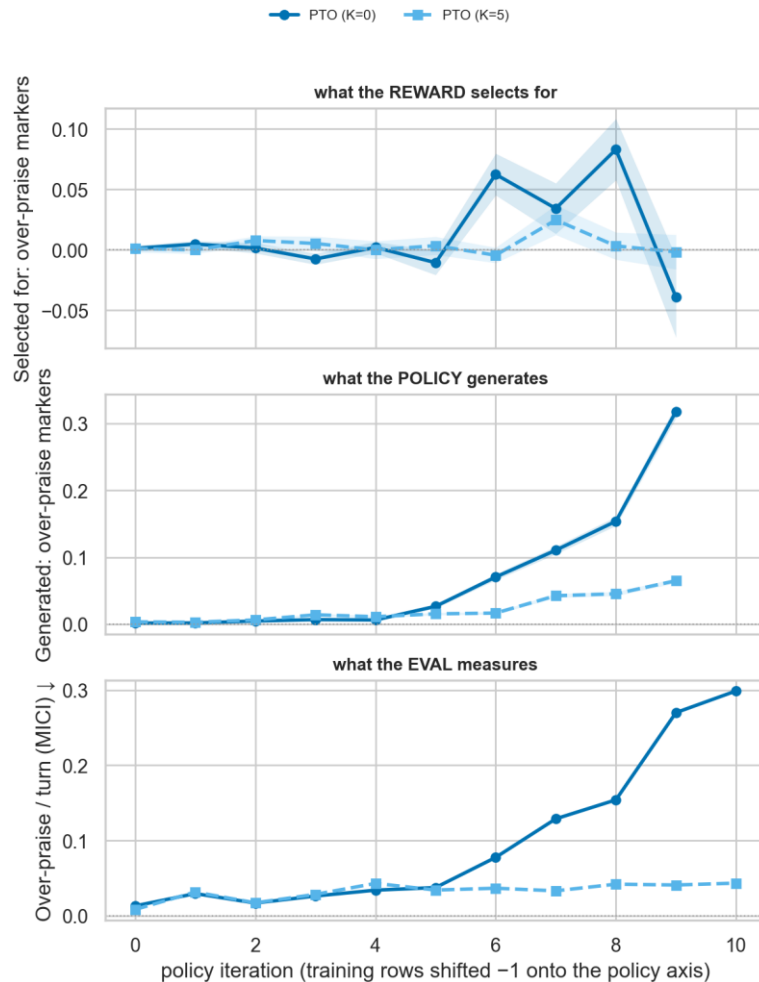
- Table above is the **primary oracle**, which WAS the training reward: totals 4.958 (K=0) vs 3.344 (K=5) acts per session.
- The held-out judge reproduces every component and **larger** — over-praise dz 0.999, advice −0.709, directing −0.468 — but its total is **null: +0.531, dz 0.099, p = .167** (8.510 vs 7.979).
- So the claim that travels across graders is **substitution, not reduction**. The significant aggregate is a primary-oracle-only result and is labelled as one wherever it appears.
- Composition flips: over-praise is **61.3%** of K=0's violations and **18.7%** of K=5's; advice-without-permission is 32.1% and **57.0%**.
- Counts, not rates.** The per-TURN MI-inconsistency rate is large under both judges (dz 0.708 / 0.655) only because K=5 takes 14.4 therapist turns to K=0's 10.2. Per-session counts are the primary reading.
- And the reward the run optimised still cannot separate the arms: Q1+Q2 dz −0.096, p_holm .695 — K=5 nominally ahead for the first time, not significantly; the held-out judge has K=0 ahead (dz 0.308, p = .032).



Per-session counts — no denominator involved.

Where the intervention acts — and why the pressure relocates

Over-praise: where look-ahead intervenes (PTO, K=0 solid · K=5 dashed)



A behaviour has to clear three gates

- Every training group logs all its candidates and their oracle scores, so we can measure what the reward **selected for** — not just what the policy produced.
- Under K=0 the selection weight on over-praise is ~ 0 for **six** iterations, then jumps — 0.063, 0.034, **0.083** at policy iterations 6–8 (SEs 0.017–0.025), before falling back. The candidate pool and the evaluation follow with a lag.
- Under K=5 it **never clears +0.025** at any iteration, and downstream stays an order smaller rather than static — the pool moves 0.004 $\rightarrow$ 0.065 and the evaluated rate 0.008 $\rightarrow$ 0.043. Look-ahead acted on the **reward**, not on the policy.
- Which is also why the substitution is unsurprising. The objective did not change; only the price of one route went up. A trajectory-extended oracle still rewards a turn that makes the next five look productive — and **giving advice does that too**. Nothing penalises advice; the intervention penalises praise that does not compound.
- Scale worth holding together: the largest per-iteration selection pressure anywhere in this chain is **0.083**, while the generated over-praise rate moves **0.002 $\rightarrow$ 0.318**. A bias too small to see in one update compounds through the on-policy loop.

Two readings this corrected — and why I am putting them on a slide

STATUS.md: "drop the look-ahead MI-consistency claim" (a summary hid its components)

Correct that TOTAL MICI flips sign across iterations 7–8 — because the total mixes channels that move in opposite directions. The over-praise component does not flip; it is monotone. The claim was dropped for a property of the summary, not of the data.

Earlier today: "look-ahead cuts the reward hack ~4x" (a channel scored as if it were the outcome)

True of over-praise, replicated by a second judge, mechanistically explained at three levels — and wrong about the therapist's behaviour. It was scored on the channel that motivated the fix. The aggregate had not been looked at.

- **Both errors have the same shape:** a claim about a summary or a component, reported as a claim about the behaviour. Neither table was ever wrong.
- **The fix is cheap and now in the analysis code:** per-session counts are tested as their own family beside the per-turn rates, and the aggregate is plotted on the same axes as the channel, so a channel result cannot be shown without its total.
- **The methodological point is the paper's contribution, not an aside.** Any intervention will look successful when it is scored on the channel that motivated it. The aggregate has to be nominated BEFORE the intervention.

Two drafts, deliberately disjoint

Affirmation Without Inquiry — $K = 0$ only (complete draft)

What the reward hack IS. PTO vs GRPO at matched look-ahead; the loop works on its own terms, what it actually taught (turns 2.3–3.4× longer, questions collapse), and how much survives the held-out judge. 8-page body, fits, 0 overfull boxes.

The Hack Moves — PTO only, $K \in \{0,5\}$ (drafted this week)

Whether a principled fix removes it. The channel closes for the whole run; the failure substitutes rather than falls; the mechanism is measured at the reward. 8 pages, 0 overfull boxes — related work is still a scaffold of citation slots.

- **They share no claims.** The split is the K axis: the first holds the optimiser as the variable and K fixed at 0; the second holds the optimiser fixed and varies K . Each carries a claim→artifact ledger, and the over-praise numbers appear in both at different iterations — the one place they could cross-contaminate.
- **Neither is a $K \times$ method result.** GRPO at $K=5$ has one iteration. Both drafts say so in Limitations rather than implying otherwise.
- **Every number in both traces to a generated artifact**, not to a summary document — which is what caught the two errors on the previous slide.

Decision 1 — scope and order

Two drafts exist and the thesis needs a spine. These are not mutually exclusive; the question is what gets finished first and what the chapter is built around.

Submit the K=0 paper first, keep look-ahead for the thesis (lowest risk)

The reward-hacking result is the more complete and the more self-contained. Look-ahead becomes a thesis chapter where the GRPO gap is a stated limitation rather than a reviewer's first question.

Submit the look-ahead paper first (higher ceiling, higher risk)

It is the more novel claim and the one with a methodological contribution beyond MI. But it rests on one optimiser and one run per arm, and a reviewer will ask for GRPO at K=5.

Merge into one longer paper

The hack, then the attempted fix, then the substitution — one arc. Does not fit 8 pages, so it means a different venue and a later date.

My read: the first option. The K=0 draft is closer to done, and the look-ahead result gets materially stronger for ~one more spend decision — which is Decision 2.

Decision 2 — what, if anything, is worth buying

Total API spend to date is roughly \$312 and is the binding constraint on everything below. Two candidates are left; only one of them changes a claim.

GRPO at K = 5, five or six iterations (changes a claim)

The only purchase that turns the look-ahead result from a within-PTO finding into a $K \times$ method result. It is also the one a reviewer will ask for. Requires training, not just scoring, so it is by far the most expensive thing on this slide.

Nothing further

Both drafts are defensible as they stand, with the gaps named in Limitations. Spend the remaining budget on nothing, and write.

PTO K = 5 out to iterations 9 and 10 (done — off the list)

Was the third candidate on this slide. It has been run: scoring both graders cost $\approx$ \$2 and closed the prevention-vs-delay question — K=5's over-praise stays low through iteration 10 under both. It did not change the substitution result, which now stands over eleven matched iterations.

Note on cost asymmetry: K=5 spent MORE than K=0 over the full ten iterations (7,548 preference groups built vs 6,240, each paying five extra simulated turns). Matching iterations is the right control, but it is generous to K=0 on budget.

What I'd like to leave with

- **A decision on order.** Which draft is finished first, and is the other a thesis chapter or a second submission?
- **A go / no-go on GRPO at $K = 5$.** It is the only remaining purchase that changes what either paper can claim.
- **A view on human MI coders.** Every behaviour count in both drafts is produced by a language model applying a coding scheme. Two graders from different families agree, but agreement is not validity, and no MI-trained human has read a single transcript. Even 40–50 coded sessions would change what we can claim — do we have access to a coder?
- **Co-author list and order** for both drafts, plus venue and target date.
- **A sanity check on the clinical reading.** Over-praise and unsolicited advice are both MI-inconsistent, but they fail differently, and our two graders disagree about which mix is more severe. I have declined to rank them — is that the right call?

Everything in this deck regenerates from one command. Figures, tables and summaries rebuild under either grader with seeded intervals; the analysis package carries a 20-check self-test that runs after every change.